Separating observer motion, bulk flow and a two-mode history by finite response design

HTT conditional response-design research

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Abstract

A radial velocity response measures the difference between source bulk motion and observer motion. A time-integrated channel also loses any history mode orthogonal to its window. We construct a finite example in which these two ambiguities can be separated explicitly. Thirteen radial responses at two radii determine twelve combinations of eighteen signed coefficients after removing a scalar calibration. Their six-dimensional kernel consists of a common observer–bulk displacement and an unweighted history mode. An independently calibrated three-component observer response removes the first ambiguity, while a second normalized history window removes the second. Either addition raises the rank to fifteen; together they give rank eighteen. Equal radii provide a negative control: the joint rank falls to fourteen. A single, fixed column normalization permits comparisons of all four designs, radial leverage and channel precision. Singular-value bounds quantify the deterministic error amplification and the response accuracy required for a stable inverse. The construction is a conditional measurement design with prescribed response functions and calibration, suitable for evaluating which additional information a physical history provider must supply.

I. A COMMON FRAME AND A FINITE RESPONSE

Consider a nonrotating orthonormal spatial frame carried by an expanding fiducial normal observer. We use signature $(-+++)$, proper radial positions rn, $n \cdot n = 1$, and the fixed scales

$$\begin{aligned} c &= 299792.458\ \mathrm{km\,s^{-1}}, & R_* &= 100\ \mathrm{Mpc}, \\ v_* &= 300\ \mathrm{km\,s^{-1}}, & \beta_* &= v_*/c, \\ h_* &= v_*/R_* = 3\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \end{aligned} \tag{1}$$

Our dimensionless radial datum is $y = (c/v_*)\delta z$. All vector components and tensors below refer to this same frame. The source bulk and observer velocities are $v_* b$ and $v_* \ell$. A positive observer velocity along the observer-to-source direction reduces the first-order radial redshift, giving the relative-motion term $n \cdot (b - \ell)$. A scalar rate coefficient $h_* m$ and a trace-free symmetric rate $h_* S$ complete the local affine part of the response. These are coefficients of a local first-order design; the reference frame and scales do not supply a finite-redshift propagation law or an absolute background expansion measurement.

Choose the Frobenius-orthonormal STF basis

$$F_1 = \frac{\mathrm{diag}(1,-1,0)}{\sqrt{2}}, \quad F_2 = \frac{\mathrm{diag}(1,1,-2)}{\sqrt{6}},$$
$$F_3 = \frac{e_x e_y^T + e_y e_x^T}{\sqrt{2}}, \qquad F_4 = \frac{e_x e_z^T + e_z e_x^T}{\sqrt{2}}, \qquad F_5 = \frac{e_y e_z^T + e_z e_y^T}{\sqrt{2}}. \tag{2}$$

Thus $S = \sum_{a=1}^{5} s_a F_a$ and $\|S\|_F^2 = \sum_a s_a^2$. We retain all five signed coordinates and the full matrix, rather than its norm alone.

Let $\tau \in [0,1]$ parametrize a stipulated two-mode matter-tilt history,

$$\beta_G(\tau) = \beta_*[g_0 + (2\tau - 1)g_1]. \tag{3}$$

The coefficients g_0, g_1 are dimensionless vectors in the transported common frame. For a normalized window W, prescribe the dimensionless sensitivity

$$\rho n \cdot \int_0^1 W(\tau) \frac{\beta_G(\tau)}{\beta_*}\, d\tau, \qquad \rho = \frac{r}{R_*}. \tag{4}$$

Equation (4) defines this experiment. It does not derive a universal relation between tilt, acceleration and a redshift-rate dipole. A physical application must provide that relation and its window independently. The illustrative coefficient box $[-1,1]^{18}$ ensures $\sup_\tau \|\beta_G(\tau)\| \le 2\sqrt{3}\beta_* < 1$; the box serves only to locate the finite examples and is not a probability distribution or a cosmological admissible region.

Order the parameters as

$$\theta = (\ell_x, \ell_y, \ell_z, b_x, b_y, b_z, m, g_{0x}, g_{0y}, g_{0z}, s_1, \ldots, s_5, g_{1x}, g_{1y}, g_{1z})^T. \tag{5}$$

For the baseline window $W_0 = 1$, the integral of $2\tau - 1$ vanishes. With one free scalar calibration α, the thirteen baseline means are

$$y_i = \alpha + n_i \cdot (b - \ell) + \rho_i[m + n_i \cdot g_0 + n_i^T S n_i]. \tag{6}$$

The first nine directions, all at $\rho = 1$, are

$$e_x, -e_x, e_y, -e_y, e_z, -e_z, (3/5, 4/5, 0), (3/5, 0, 4/5), (0, 3/5, 4/5). \tag{7}$$

Append $e_x, -e_x, e_y, e_z$ at $\rho = q > 0$. The nominal design uses $q = 2$. Define E_3 to have rows n_i^T, define Q_5 to have rows $(n_i^T F_a n_i)_{a=1}^5$, and set $R = \mathrm{diag}(\rho_i)$. The baseline design and nuisance column are

$$J_0 = [-E_3,\ E_3,\ R\mathbf{1},\ RE_3,\ RQ_5,\ 0_{13\times 3}], \qquad N_0 = \mathbf{1}_{13}. \tag{8}$$

Here $R\mathbf{1}$ is a vector of dimensionless radii; it is distinct from the physical length R_*.

II. ONE NUISANCE QUOTIENT AND TWO ADDITIONAL RESPONSES

Two independently calibrated three-component responses are proposed:

$$z_L = \ell + e_L, \qquad z_G = \int_0^1 2\tau\, \frac{\beta_G(\tau)}{\beta_*}\, d\tau + e_G = g_0 + \frac{g_1}{3} + e_G. \tag{9}$$

The first supplies an observer-velocity anchor. The second uses the normalized window $W_1 = 2\tau$, whose overlap with $2\tau - 1$ is $1/3$. Their declared gains express both in dimensionless y-equivalent units. This is a pair of hypothetical measurement functionals, rather than a claimed delivered survey channel. Their calibration is independent in the algebraic sense that neither carries the free α of (6). No stochastic independence is assumed.

The augmented rows are

$$J_L = [I_3, 0, 0, 0, 0, 0], \qquad J_G = [0, 0, 0, I_3, 0, I_3/3], \qquad J = \begin{pmatrix} J_0 \\ J_L \\ J_G \end{pmatrix}, \qquad N = \begin{pmatrix} \mathbf{1}_{13} \\ 0_3 \\ 0_3 \end{pmatrix}. \tag{10}$$

Block widths in (10) follow (5). Removing an addition deletes its three rows from both J and N. For every design, the scalar nuisance is removed once, using the complete retained data vector and the same construction for all parameter columns. For a positive row-precision matrix W, define

$$P = I - \frac{(WN)(WN)^T}{\|WN\|^2}, \qquad A_{\text{phys}} = PWJ, \qquad \bar{y} = PWy. \tag{11}$$

For the full design we take

$$W = \operatorname{diag}(I_{13}, \sigma_L^{-1} I_3, \sigma_G^{-1} I_3), \tag{12}$$

with nominal $\sigma_L = \sigma_G = 1$. The subdesigns retain the applicable diagonal blocks. These positive scales define a deterministic comparison metric. A physical joint sampling covariance

and a statistical law are not supplied. In particular, the singular-value plots below do not carry confidence levels.

The kernel characterization

$$PWJ\theta = 0 \quad \Longleftrightarrow \quad J\theta \in \operatorname{col} N \tag{13}$$

holds because W is invertible and P projects away exactly WN. Thus finite positive precision weights alter conditioning but not structural identifiability. Allowing unknown offsets for the added responses would enlarge N and require a new analysis.

III. EXACT RANKS AND IDENTIFIED COMBINATIONS

Proposition 1 (two-radius separation). For (7), the nine-dimensional spherical polynomial space consisting of a monopole, a dipole and an STF quadrupole is unisolvent. With the four repeated directions in (7) placed at any distinct positive radius $q \neq 1$, the matrix

$$X = [E_4, RE_9], \qquad E_4 = [\mathbf{1}, E_3], \quad E_9 = [\mathbf{1}, E_3, Q_5] \tag{14}$$

has rank thirteen.

Proof. Let $f(n) = a + d \cdot n + n^T T n$, T STF, vanish on the first nine directions. The opposite axis pairs first give $d = 0$, then $a + T_{ii} = 0$. Taking the trace gives $a = 0$ and all three diagonal entries vanish. At the three remaining directions the nonzero cross terms force $T_{xy} = T_{xz} = T_{yz} = 0$. Hence E_9 is invertible. A degree-one polynomial vanishing at $e_x, -e_x, e_y, e_z$ is likewise zero, so the repeated-direction E_4 is invertible. A null vector of (14) represents polynomials $p_1 + \rho p_2$. At $\rho = 1$, unisolvence gives $p_2 = -p_1$ as polynomials. At the four repeated directions the equation becomes $(1 - q)p_1 = 0$, forcing $p_1 = p_2 = 0$. This is the retained two-radius rank argument, expressed in the orthonormal tensor coordinates (2). □

Proposition 2 (baseline quotient). At $q \neq 1$, the baseline response after (11) has rank twelve. Its exact six-dimensional kernel is

$$k_{\text{split}}(v) = (v, v, 0, 0, 0, 0), \qquad k_{\text{hist}}(w) = (0, 0, 0, 0, 0, w), \qquad v, w \in \mathbb{R}^3. \tag{15}$$

The identified signed combinations are

$$b-\ell, \qquad m, \qquad g_0, \qquad (s_1,\dots,s_5). \tag{16}$$

Proof. By (13), a quotient-null vector satisfies $J_0\theta = a\mathbf{1}$ for some a. Proposition 1 applied to the coefficients $(-a, b-\ell, m, g_0, s)$ forces $a=0$, $b-\ell=0$, $m=g_0=s=0$. No row contains g_1. These are precisely (15), proving both completeness and the rank. □

The ambiguity in (15) has two separate mechanisms. A shared change to source bulk and observer velocity leaves their relative radial velocity unchanged. A history proportional to $2\tau-1$ has zero unweighted mean. Repeated angular or radial observations with the same response window cannot reveal that history coefficient.

Proposition 3 (constructive augmentation). At $q \neq 1$, either response in (9) gives quotient rank fifteen and a three-dimensional kernel. Their joint addition gives quotient rank eighteen and a unique eighteen-coordinate inverse on the quotient data image.

Proof. The first addition acts on (15) as $J_L k_{\mathrm{split}}(v) = v$ and $J_L k_{\mathrm{hist}}(w) = 0$. The second acts as $J_G k_{\mathrm{split}}(v) = 0$ and $J_G k_{\mathrm{hist}}(w) = w/3$. Hence each addition removes exactly its corresponding three-dimensional kernel, and together they remove all six dimensions. Equation (13) extends the result to every positive finite precision weighting in (12). □

This count separates information supplied by an endpoint anchor from information supplied by a new history moment. The same conclusion follows by reconstructing

$$\ell = z_L, \qquad b = (b-\ell) + z_L, \qquad g_1 = 3(z_G - g_0), \tag{17}$$

after solving for (16).

Proposition 4 (equal-radius negative control). At $q=1$, the baseline quotient has rank eight. Either addition has rank eleven, and their joint addition has rank fourteen.

Proof. At one radius, (6) depends only on the scalar $\alpha+m$, the dipole $b-\ell+g_0$, and the five STF coefficients. The nuisance quotient removes the scalar, leaving the independent dipole and quadrupole, hence rank eight. Each calibrated addition contributes three independent output coordinates. With both additions the remaining kernel is generated by

$$\delta\theta_d = (0,-d,0,d,0,-3d), \qquad \delta\theta_m = (0,0,t,0,0,0), \quad d\in\mathbb{R}^3, t\in\mathbb{R}. \tag{18}$$

The first leaves every raw mean, including both additions, unchanged. The second is absorbed by $\delta\alpha = -t$. Conversely, a joint quotient-null vector obeys $\ell=0$, $g_1=-3g_0$,

$b = -g_0$ and $s = 0$, with m free. Thus the kernel has exactly four dimensions. □

Equation (18) exhibits why independent endpoint and history anchors do not replace radial leverage. Distinct radii separate an angular intercept from a depth-proportional dipole and monopole.

IV. EXPLICIT FINITE WITNESSES

Starting from $\theta = 0$, choose $v = e_x/4$ and $w = e_y/4$. The two states $k_{\mathrm{split}}(v)$ and $k_{\mathrm{hist}}(w)$ both remain in the illustrative box. Every baseline prediction is exactly zero for each state. The first changes z_L by $(1/4, 0, 0)$; the second changes z_G by $(0, 1/12, 0)$. Their sum changes both additions while leaving all thirteen baseline means unchanged. These signed nineteen-coordinate predicted arrays are retained in the numerical supplement.

The history witness corresponds to

$$\beta_G(\tau) = \frac{\beta_*}{4}(2\tau - 1)e_y. \tag{19}$$

Its time integral vanishes although the history is nonzero except at its midpoint. The zero baseline column is therefore a response null direction, not a statement that the physical history vanishes.

For an equal-radius joint witness in the same box, take $d = e_x/4$ in (18): $b = -e_x/4$, $g_0 = e_x/4$, $g_1 = -3e_x/4$, with all other coordinates zero. All nineteen raw means remain zero. A separate change $m = 1/4$, $\alpha = -1/4$ supplies the monopole witness.

V. FIXED NORMALIZATION AND SPECTRAL COMPARISON

Velocity and rate scales in (1) are specified before numerical normalization. Let J_* and P_* denote the full design at $q = 2$, $\sigma_L = \sigma_G = 1$. Define once

$$d_j = \|P_* J_* e_j\|_2 > 0, \qquad L = \mathrm{diag}(d_1^{-1}, \ldots, d_{18}^{-1}), \qquad \theta = Lz. \tag{20}$$

The normalized response is

$$A = PWJL. \tag{21}$$

Nominal augmented columns have unit norm. The same numerical matrix L is then frozen for every subdesign, radius and precision case. In particular, the baseline's three zero history columns remain zero; weaker channels are not restored to unit norm during a scan. The supplement records the complete d_j, L, unscaled and scaled spectra and all physical unit conversions.

With $A = U\Sigma V^T$, the identified scaled coordinates are $V_r^T z = V_r^T L^{-1}\theta$, where V_r contains the right singular vectors for the nonzero spectrum. The corresponding projector is $\Pi_{\mathrm{id}} = V_r V_r^T$, and the kernel projector is $I - \Pi_{\mathrm{id}}$. The named physical combinations (16) span the same identified information but are not an orthonormal singular-vector basis. Singular-vector signs and bases within repeated singular subspaces have no physical significance.

The baseline is a 13×18 matrix of rank twelve. A thin SVD returns thirteen singular values and therefore only one explicit zero. The parameter kernel still has six dimensions. The complete eighteen-dimensional right basis and kernel projector are required to retain the five additional null directions absent from that thin list. The plotted spectrum may pad the missing five entries with structural zeros provided the caption identifies the padding.

The finite comparison uses the following analytic targets:

Design	Rows	Rank at $q = 2$	Kernel dimension	Rank at $q = 1$
Baseline	13	12	6	8
Baseline plus endpoint	16	15	3	11
Baseline plus history	16	15	3	11
Both additions	19	18	0	14

Design	Rank	Kernel	Positive smin	Identified condition	Full-vector gain
baseline	12	6	0.10665269	16.950054	unbounded
endpoint	15	3	0.097638737	18.655425	unbounded
history	15	3	0.10180602	17.900956	unbounded
both	18	0	0.093938751	19.539575	10.6452

To show radial sensitivity, set $q = 1 + \delta$ for

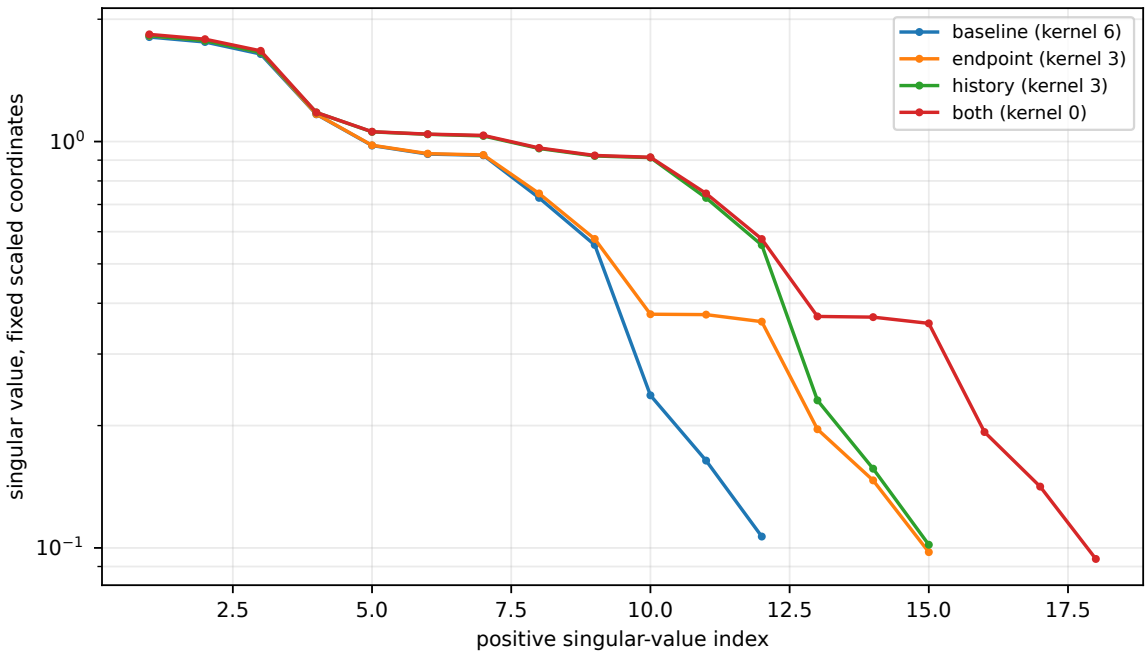


FIG. 1. Singular spectra at q=2 with the fixed normalization (20). Legends state structural null dimensions. The row metric is stipulated, not measured covariance.

$$\delta \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 3\}. \tag{22}$$

At every positive separation, the analytic ranks are those of Proposition 3; as the separation approaches zero, four joint singular directions become weak and at equality join the exact kernel. Numerical threshold dependence near equality should be displayed alongside the structural rank. A finite threshold does not change the algebraic theorem. At equality the inverse gain for all eighteen coordinates is infinite, even though the fourteen identified directions retain finite gains.

For additional-channel precision, evaluate every pair

$$(\sigma_L, \sigma_G) \in \{1/2, 1, 2, 4\}^2 \tag{23}$$

at $q = 2$, preserving (20). These cases measure how the weakest recoverable combination responds to the relative precision of the two added channels.

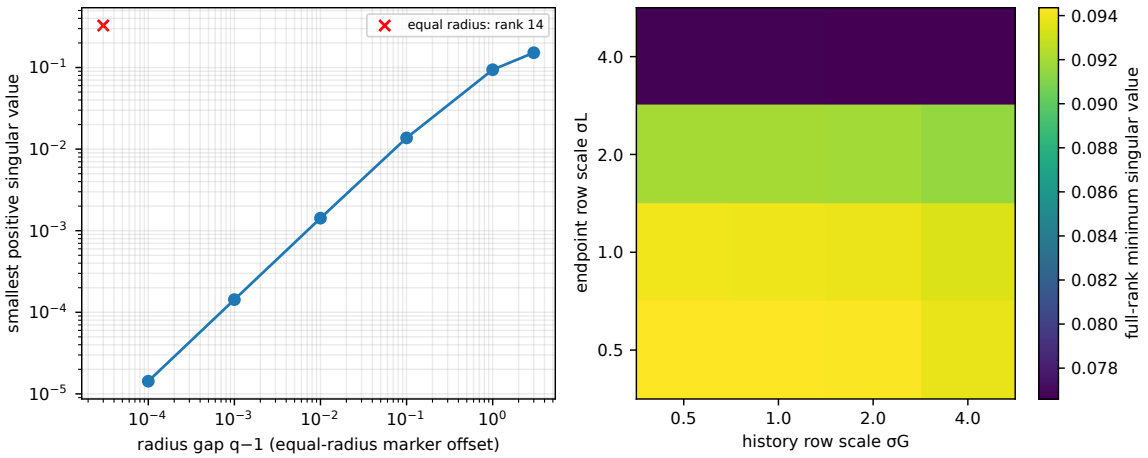


FIG. 2. Radial leverage and additional-channel precision in fixed scaled coordinates. The equal-radius marker is offset horizontally for log display; its full-vector inverse gain is unbounded.

VI. DETERMINISTIC ERROR SENSITIVITY

Let the normalized projected data obey $\bar{y} = Az + e$, with $\|e\|_2 \le \epsilon$. Suppose the operator used for inversion is $\widetilde{A} = A + \Delta$, where $\|\Delta\|_2 \le \eta$. Both errors are measured after the prescribed quotient and normalization. Thus Δ must include any error in that effective operator; nuisance-model changes cannot be treated as harmless coordinate changes. For a full-column-rank A, Weyl's bound gives

$$s_{\min}(\widetilde{A}) \ge s_{\min}(A) - \eta. \tag{24}$$

If $s_{\min}(A) > \eta$, the least-squares inverse satisfies

$$\|\widehat{z} - z\|_2 \le \frac{\epsilon + \eta\|z\|_2}{s_{\min}(A) - \eta}. \tag{25}$$

Proof. Full column rank of $\widetilde{A}$ gives $\widehat{z} - z = \widetilde{A}^{+}(e - \Delta z)$. Taking norms and using (24) proves (25). No distribution or independence assumption is used. □

The physical coefficient error obeys $\|\widehat{\theta} - \theta\|_2 \le \|L\|_2\|\widehat{z} - z\|_2$, and any specified linear physical functional can use its own sharper factor $\|CL\|_2$. The illustration $\epsilon = 10^{-3}$, $\|z\|_2 \le 1$ is a separate error scenario, not the claim that the whole box $[-1, 1]^{18}$ lies in the unit z-ball. For $\eta/s_{\min} \in \{0, 0.1, 0.25, 0.5, 0.75, 0.9, 1, 1.1\}$, the bound loses certification at one. Such cases are reported as `NOT_CERTIFIED_BY_THIS_BOUND`.

For a fixed rank-deficient operator, the minimum-norm estimate recovers the orthogonal identified projection with

$$\|\widehat{z} - \Pi_{\mathrm{id}} z\|_2 \leq \frac{\epsilon}{s^+_{\min}(A)}, \tag{26}$$

where $s^+_{\min}$ is the smallest nonzero singular value. This controls identified combinations only. An arbitrary kernel component leaves the data unchanged, so no uniform full-vector bound follows. Perturbations that change rank or the kernel require their own subspace control; (26) is the fixed-operator statement.

Equation (17) also gives direct, physically transparent bounds. If the errors on the estimated combinations obey $\|\delta(b - \ell)\| \leq e_D$, $\|\delta z_L\| \leq e_L$, $\|\delta g_0\| \leq e_0$, and $\|\delta z_G\| \leq e_G$, then

$$\|\delta b\| \leq e_D + e_L, \qquad \|\delta g_1\| \leq 3(e_G + e_0). \tag{27}$$

The history inverse incurs a factor of three because the new window overlaps the previously invisible temporal mode by only 1/3. This penalty survives the fact that the joint design is injective.

VII. PHYSICAL INTERPRETATION AND THE NEXT MEASUREMENT

The design identifies what must change when a baseline response cannot separate physical mechanisms. The observer–bulk ambiguity requires a response to observer motion that is not a response to the same source bulk displacement. The history ambiguity requires a window with nonzero overlap with a temporal mode invisible to the existing window. These requirements are independent of the number of rows carrying the unchanged baseline response. Radial leverage performs a third job: separating angular intercepts from rate and mean-history coefficients.

For a physical implementation, the carrier remains the signed vectors (ℓ, b, g_0, g_1), the full STF tensor S, scalar m, response functions and their joint error set in one frame. The radial response contains no antisymmetric velocity-gradient measurement, because $n^T \Omega n = 0$. Matter vorticity therefore remains `NOT_MEASURED`; a chosen hypersurface normal's irrotationality is a distinct statement. The scalar rate and fitted STF coefficients in this design do not alone provide a validated physical expansion/shear bridge.

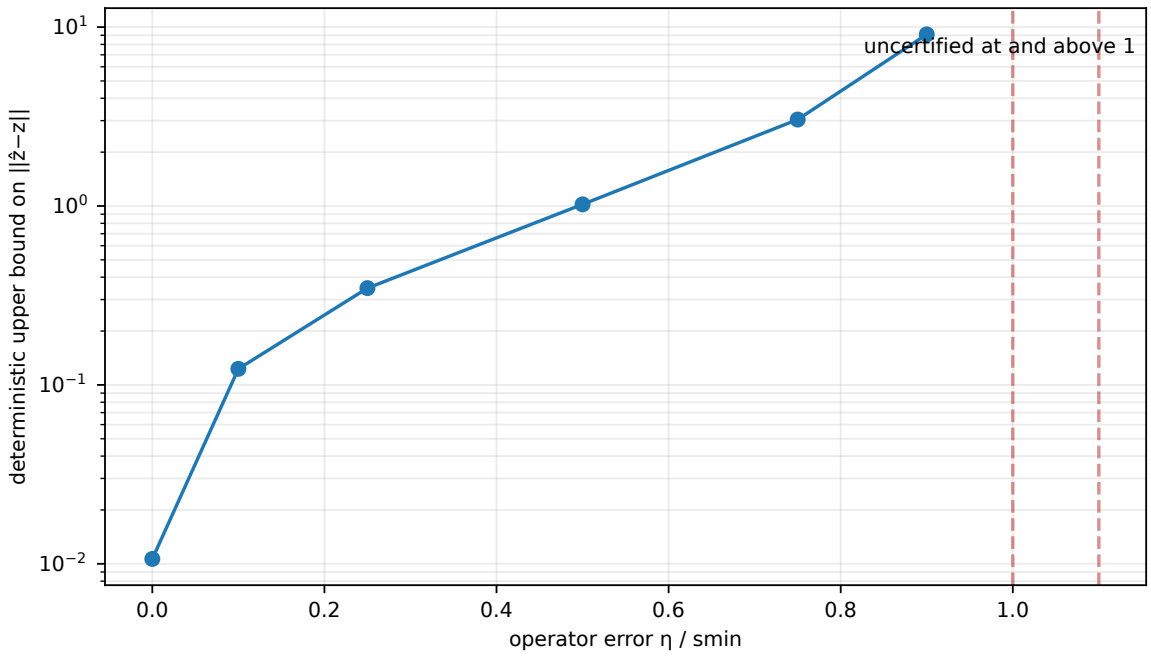


FIG. 3. Deterministic inverse bound (25) for epsilon=0.001 and ||z||<=1. At and above eta/smin=1 the bound is uncertified. No sampling law is supplied.

No map from this finite carrier to a physical signed comparator x_C, an anchored stress score Q, an occupancy F, an exceedance Π, or a depth functional G_F has been specified. Each remains `NOT_EVALUABLE_WITH_THIS_RESPONSE`. Respectively, evaluation would require the full physical comparator components and frame, a sign/domain-clean anchor, a support measure and ceiling, a joint probability law, and compatible depth supports and joint dependence. These missing definitions are retained as undefined sectors rather than filled by zero.

The original V03 external-history validation remains a separate requirement. It must supply actual endpoint and history response kernels, calibration/nuisance coupling, common-frame transport and units, and a justified response error or covariance. The result here is the exact finite conditional design and its inverse sensitivity. It neither changes the original scientific acceptance nor supplies a native transfer or family-identification result.

Appendix A: Numerical appendix: reproducible comparison

The companion consumer verifies the sealed retained inputs, constructs (2), (7)–(12), freezes (20), and emits all designs, full right singular bases, kernel/identified projectors, spectra and signed witnesses. The nominal rank counts must agree at relative cutoffs $10^{-12}, 10^{-10}, 10^{-8}$. Frobenius-basis and null residuals must be at most 10^{-12}; reconstruction and deterministic bound checks use relative tolerance 10^{-10} or absolute tolerance 10^{-12} for unit-scale fixtures. Exact analytic rank remains authoritative near equal radii. The generated tables and figures report both numerical values and their design assumptions.

The source rank theorem is retained from `mathematical_recovery.md`, "Two radii separate angular nuisance and depth response", corroborated in `independent_review.md`, "Rank-thirteen example and recovered physical scope", and related to the oracle rank criterion in `NATIVE_INFERENCE_THEOREMS.md`, §6. The endpoint/history distinction and common nuisance quotient are developed in `theory_synthesis_20260913/REPORT.md`, §§13–13.1, and `theorem_physical_chapter.md`, the response quotient and anchor-coordinate paragraphs. Their precise source identities and the new finite construction are separated in the companion claim map.

Appendix B: Generated numerical appendix tables

Parameter	Physical coefficient unit	d_j	L_jj
ell_x	v*=300 km/s	2.368381466	0.4222292795
ell_y	v*=300 km/s	2.134695078	0.4684509794
ell_z	v*=300 km/s	2.181742423	0.4583492485
b_x	v*=300 km/s	2.146911915	0.4657852952
b_y	v*=300 km/s	1.885980667	0.5302281287
b_z	v*=300 km/s	1.939071943	0.5157106231
m	h*=3 km/s/Mpc	1.664100589	0.6009252126
g0_x	beta*=300/299792.458	3.407232127	0.2934933584
g0_y	beta*=300/299792.458	2.666602563	0.3750090147
g0_z	beta*=300/299792.458	2.698717644	0.3705463601

Parameter	Physical coefficient unit	d_j	L_jj
s_1	h*=3 km/s/Mpc	2.8381359	0.3523439452
s_2	h*=3 km/s/Mpc	2.664247621	0.3753404872
s_3	h*=3 km/s/Mpc	0.6521915717	1.533291817
s_4	h*=3 km/s/Mpc	0.6521915717	1.533291817
s_5	h*=3 km/s/Mpc	0.6521915717	1.533291817
g1_x	beta*=300/299792.458	0.3333333333	3
g1_y	beta*=300/299792.458	0.3333333333	3
g1_z	beta*=300/299792.458	0.3333333333	3

1. Complete nominal scaled spectrum: baseline

Index	Singular value
1	1.80776894878
2	1.75783553205
3	1.64202991241
4	1.16693241497
5	0.977608148968
6	0.93153934285
7	0.925000426039
8	0.72695402196
9	0.556681575992
10	0.237422793074
11	0.164010989736
12	0.106652693887
13	2.28393497472e-16
14	structural zero (right completion)
15	structural zero (right completion)
16	structural zero (right completion)
17	structural zero (right completion)

Index	Singular value
18	structural zero (right completion)

2. Complete nominal scaled spectrum: endpoint

Index	Singular value
1	1.8214921072
2	1.77388823641
3	1.6579048791
4	1.16814162944
5	0.979221208186
6	0.933867897306
7	0.92679745181
8	0.745208194267
9	0.57589372613
10	0.375920041292
11	0.375052692147
12	0.360409875835
13	0.195900842657
14	0.146618339872
15	0.097638736537
16	0
17	structural zero (right completion)
18	structural zero (right completion)

3. Complete nominal scaled spectrum: history

Index	Singular value
1	1.8224251976

Index	Singular value
2	1.77075892464
3	1.6569969902
4	1.17889465675
5	1.05629533648
6	1.04195741167
7	1.03219924361
8	0.961460443345
9	0.921463617644
10	0.91312580769
11	0.726700364284
12	0.55634543484
13	0.230754137695
14	0.1566639275
15	0.101806023814
16	1.84960232996e-16
17	structural zero (right completion)
18	structural zero (right completion)

4. Complete nominal scaled spectrum: both

Index	Singular value
1	1.83552330269
2	1.78615342167
3	1.6720014412
4	1.17933809547
5	1.05743023017
6	1.04292085192
7	1.03553442499
8	0.964471288015

Index	Singular value
9	0.924256501433
10	0.914865452434
11	0.745162919861
12	0.57555461857
13	0.371204892219
14	0.36975411719
15	0.356918568169
16	0.192824411934
17	0.141550104788
18	0.0939387506842

All arrays, complete right bases and source-linked checks are in the companion JSON artifacts. This numerical appendix is generated from the frozen design and does not add observed channels.